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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Motion in a Plane

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	83 questions
Numerical	20 questions
Answer key	at the end

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MCQ (Single Correct Answer) 83 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Two identical bodies, projected with the same speed at two different angles cover the same horizontal range R. If the time of flight of these bodies are 5 s and 10 s , respectively, then the value of R is

____ m. (Take $g = 10\ \mathrm{m/s^2}$)

(A) 250 (B) 25 (C) 500 (D) 125

Open solution

Q2. JEE Main 2026 (Online) 4th April Evening Shift

If x and y coordinates of a projectile as a function of time (t) are given as $24t$ and $43.6t - 4.9t^2$, respectively, then the angle (in degrees) made by the projectile with horizontal when $t = 2\ \mathrm{s}$ is ____ .

(A) 60 (B) 45 (C) 30 (D) 75

Open solution

Q3. JEE Main 2026 (Online) 4th April Morning Shift

The two projectiles are projected with the same initial velocities at the 15° and 30° with respect to the horizontal. The ratio of their ranges is $1 : x$. The value of x is

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) $2\sqrt{3}$ (D) $\frac{1}{\sqrt{2}}$

Open solution

Q4. JEE Main 2026 (Online) 24th January Morning Shift

A boy throws a ball into air at 45° from the horizontal to land it on a roof of a building of height H. If the ball attains maximum height in 2 s and lands on the building in 3 s after launch, then value of H is ____ m.

$\left(\mathrm{g} = 10\ \mathrm{m/s^2}\right)$

(A) 20 (B) 25 (C) 10 (D) 15

Open solution

Q5. JEE Main 2026 (Online) 22nd January Morning Shift

A projectile is thrown upward at an angle 60° with the horizontal. The speed of the projectile is $20\ \mathrm{m/s}$ when its direction of motion is 45° with the horizontal. The initial speed of the projectile is ____ $\mathrm{m/s}$.

(A) $20\sqrt{3}$ (B) $20\sqrt{2}$ (C) 40 (D) $40\sqrt{2}$

Open solution

Q6. JEE Main 2026 (Online) 21st January Evening Shift

A river of width 200 m is flowing from west to east with a speed of 18 km/h. A boat, moving with speed of 36 km/h in still water, is made to travel one-round trip (bank to bank of the river). Minimum time taken by the boat for this journey and also the displacement along the river bank are _____ and _____ respectively.

(A) 20 s and 100 m (B) 40 s and 100 m (C) 40 s and 200 m (D) 40 s and 0 m

Open solution

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Q7. JEE Main 2025 (Online) 8th April Evening Shift

Two balls with same mass and initial velocity, are projected at different angles in such a way that maximum height reached by first ball is 8 times higher than that of the second ball. T_1 and T_2 are the total flying times of first and second ball, respectively, then the ratio of T_1 and T_2 is

Open solution

(A) 2 : 1 (B) $\sqrt{2}:1$ (C) $2\sqrt{2}:1$ (D) 4 : 1

Q8. JEE Main 2025 (Online) 7th April Evening Shift

A helicopter flying horizontally with a speed of 360 km/h at an altitude of 2 km, drops an object at an instant. The object hits the ground at a point O, 20 s after it is dropped. Displacement of 'O' from the position of helicopter where the object was released is :

(use acceleration due to gravity g = 10 m/s^2 and neglect air resistance)

Open solution

(A) 7.2 km (B) $2\sqrt{5}$ km (C) $2\sqrt{2}$ km (D) 4 km

Q9. JEE Main 2025 (Online) 7th April Morning Shift

Two projectiles are fired from ground with same initial speeds from same point at angles $(45^\circ+\alpha)$ and $(45^\circ-\alpha)$ with horizontal direction. The ratio of their times of flights is

Open solution

(A) $\frac{1+\tan\alpha}{1-\tan\alpha}$ (B) $\frac{1+\sin 2\alpha}{1-\sin 2\alpha}$

(C) $\frac{1-\tan\alpha}{1+\tan\alpha}$ (D) 1

Q10. JEE Main 2025 (Online) 3rd April Evening Shift

A particle is projected with velocity u so that its horizontal range is three times the maximum height attained by it. The horizontal range of the projectile is given as $\frac{nu^2}{25g}$, where value of n is: (Given, ' g ' is the acceleration due to gravity.)

Open solution

(A) 6 (B) 12 (C) 18 (D) 24

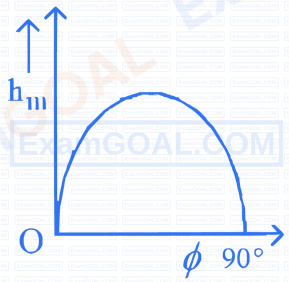
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Q11. JEE Main 2025 (Online) 3rd April Morning Shift

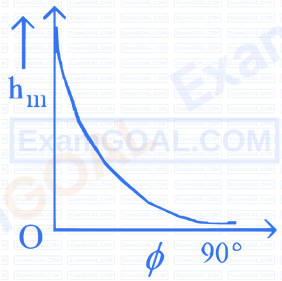
Open solution

The angle of projection of a particle is measured from the vertical axis as ϕ and the maximum height reached by the particle is h_m. Here h_m as function of ϕ can be presented as

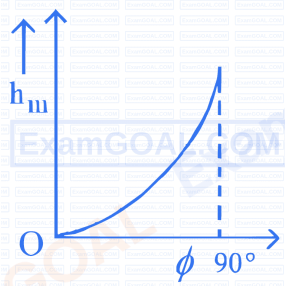
(A)



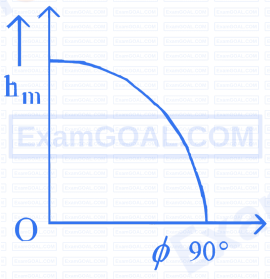
(B)



(C)



(D)



Q12. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

A river is flowing from west to east direction with speed of $9\ \mathrm{km\ h^{-1}}$. If a boat capable of moving at a maximum speed of $27\ \mathrm{km\ h^{-1}}$ in still water, crosses the river in half a minute, while moving with maximum speed at an angle of 150° to direction of river flow, then the width of the river is :

(A) 112.5 m

(B) 75 m

(C) 300 m

(D) $112.5 \times \sqrt{3}\ \mathrm{m}$

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Q13. JEE Main 2025 (Online) 29th January Morning Shift

Two projectiles are fired with same initial speed from same point on ground at angles of $(45^\circ - \alpha)$ and $(45^\circ + \alpha)$, respectively, with the horizontal direction. The ratio of their maximum heights attained is :

(A) $\frac{1+\sin\alpha}{1-\sin\alpha}$
(B) $\frac{1+\sin 2\alpha}{1-\sin 2\alpha}$
(C) $\frac{1-\tan\alpha}{1+\tan\alpha}$
(D) $\frac{1-\sin 2\alpha}{1+\sin 2\alpha}$

Open solution

Q14. JEE Main 2025 (Online) 24th January Evening Shift

The position vector of a moving body at any instant of time is given as $\vec{r} = \left(5t^2\hat{i} - 5t\hat{j}\right)$m. The magnitude and direction of velocity at $t = 2s$ is,

(A) $5\sqrt{17}\,\mathrm{m/s}$, making an angle of $\tan^{-1}4$ with - ve Y axis
(B) $5\sqrt{15}\,\mathrm{m/s}$, making an angle of $\tan^{-1}4$ with + ve X axis
(C) $5\sqrt{17}\,\mathrm{m/s}$, making an angle of $\tan^{-1}4$ with + ve X axis
(D) $5\sqrt{15}\,\mathrm{m/s}$, making an angle of $\tan^{-1}4$ with $-$ ve Y axis

Open solution

Q15. JEE Main 2025 (Online) 23rd January Evening Shift

A ball having kinetic energy KE, is projected at an angle of 60° from the horizontal. What will be the kinetic energy of ball at the highest point of its flight?

(A) $\frac{(\mathrm{KE})}{2}$
(B) $\frac{(\mathrm{KE})}{8}$
(C) $\frac{(\mathrm{KE})}{4}$
(D) $\frac{(\mathrm{KE})}{16}$

Open solution

Q16. JEE Main 2025 (Online) 22nd January Evening Shift

A ball of mass 100 g is projected with velocity $20\,\mathrm{m/s}$ at 60° with horizontal. The decrease in kinetic energy of the ball during the motion from point of projection to highest point is

(A) 20 J
(B) 5 J
(C) 15 J
(D) zero

Open solution

Q17. JEE Main 2024 (Online) 8th April Evening Shift

The angle of projection for a projectile to have same horizontal range and maximum height is :

(A) $\tan^{-1}\left(\frac{1}{2}\right)$
(B) $\tan^{-1}(2)$
(C) $\tan^{-1}\left(\frac{1}{4}\right)$
(D) $\tan^{-1}(4)$

Open solution

Q18. JEE Main 2024 (Online) 4th April Morning Shift

The co-ordinates of a particle moving in x-y plane are given by : $x = 2 + 4t, y = 3t + 8t^2$.

The motion of the particle is :

(A) uniform motion along a straight line.
(B) non-uniformly accelerated.
(C) uniformly accelerated having motion along a straight line.
(D) uniformly accelerated having motion along a parabolic path.

Open solution

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Q19. JEE Main 2024 (Online) 1st February Morning Shift

A particle moving in a circle of radius R with uniform speed takes time T to complete one revolution.

If this particle is projected with the same speed at an angle θ to the horizontal, the maximum height attained by it is equal to $4R$. The angle of projection θ is then given by :

(A) $\sin^{-1}\left[\frac{2gT^2}{\pi^2R}\right]^{\frac{1}{2}}$

(B) $\sin^{-1}\left[\frac{\pi^2R}{2gT^2}\right]^{\frac{1}{2}}$

(C) $\cos^{-1}\left[\frac{\pi R}{2gT^2}\right]^{\frac{1}{2}}$

(D) $\cos^{-1}\left[\frac{2gT^2}{\pi^2R}\right]^{\frac{1}{2}}$

Open solution

Q20. JEE Main 2024 (Online) 30th January Evening Shift

Projectiles A and B are thrown at angles of 45° and 60° with vertical respectively from top of a $400\ \mathrm{m}$ high tower. If their ranges and times of flight are same, the ratio of their speeds of projection $v_A : v_B$ is :

[Take $g = 10\ \mathrm{ms}^{-2}$]

(A) $1:2$

(B) $\sqrt{2}:1$

(C) $1:\sqrt{2}$

(D) $1:\sqrt{3}$

Open solution

Q21. JEE Main 2024 (Online) 27th January Morning Shift

Position of an ant (S in metres) moving in Y-Z plane is given by $S = 2t^2\hat{j} + 5\hat{k}$ (where t is in second). The magnitude and direction of velocity of the ant at $\mathrm{t} = 1\ \mathrm{s}$ will be :

(A) $16\ \mathrm{m/s}$ in y-direction

(B) $4\ \mathrm{m/s}$ in x-direction

(C) $9\ \mathrm{m/s}$ in z-direction

(D) $4\ \mathrm{m/s}$ in y-direction

Open solution

Q22. JEE Main 2023 (Online) 11th April Evening Shift

A projectile is projected at 30° from horizontal with initial velocity $40\ \mathrm{ms}^{-1}$. The velocity of the projectile at $\mathrm{t} = 2\ \mathrm{s}$ from the start will be : (Given $g = 10\ \mathrm{m/s}^2$)

(A) $20\sqrt{3}\ \mathrm{ms}^{-1}$

(B) Zero

(C) $20\ \mathrm{ms}^{-1}$

(D) $40\sqrt{3}\ \mathrm{ms}^{-1}$

Open solution

Q23. JEE Main 2023 (Online) 10th April Evening Shift

Two projectiles are projected at 30° and 60° with the horizontal with the same speed. The ratio of the maximum height attained by the two projectiles respectively is:

(A) $1:\sqrt{3}$

(B) $\sqrt{3}:1$

(C) $1:3$

(D) $2:\sqrt{3}$

Open solution

Q24. JEE Main 2023 (Online) 10th April Morning Shift

The range of the projectile projected at an angle of 15° with horizontal is 50 m. If the projectile is projected with same velocity at an angle of 45° with horizontal, then its range will be

(A) $50\sqrt{2}$ m

(B) 100 m

(C) $100\sqrt{2}$ m

(D) 50 m

Open solution

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Q25. JEE Main 2023 (Online) 8th April Evening Shift

The trajectory of projectile, projected from the ground is given by $y = x - \frac{x^2}{20}$. Where x and y are measured in meter. The maximum height attained by the projectile will be.

(A) 10 m (B) 5 m (C) 200 m (D) $10\sqrt{2}$ m

Open solution

Q26. JEE Main 2023 (Online) 8th April Morning Shift

Two projectiles A and B are thrown with initial velocities of $40\ \mathrm{m/s}$ and $60\ \mathrm{m/s}$ at angles 30° and 60° with the horizontal respectively. The ratio of their ranges respectively is $\left(g = 10\ \mathrm{m/s^2}\right)$

(A) $4 : 9$ (B) $2 : \sqrt{3}$ (C) $\sqrt{3} : 2$ (D) $1 : 1$

Open solution

Q27. JEE Main 2023 (Online) 6th April Morning Shift

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : When a body is projected at an angle 45°, it's range is maximum.

Reason R : For maximum range, the value of $\sin 2\theta$ should be equal to one.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) $\mathbf{A}$ is true but $\mathbf{R}$ is false

(C) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are correct but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

Open solution

Q28. JEE Main 2023 (Online) 1st February Evening Shift

For a body projected at an angle with the horizontal from the ground, choose the correct statement.

(A) Gravitational potential energy is maximum at the highest point.

(B) The vertical component of momentum is maximum at the highest point.

(C) The horizontal component of velocity is zero at the highest point.

(D) The Kinetic Energy (K.E.) is zero at the highest point of projectile motion.

Open solution

Q29. JEE Main 2023 (Online) 1st February Morning Shift

A child stands on the edge of the cliff $10\ \mathrm{m}$ above the ground and throws a stone horizontally with an initial speed of $5\ \mathrm{ms^{-1}}$. Neglecting the air resistance, the speed with which the stone hits the ground will be _____ $\mathrm{ms^{-1}}$ (given, $g = 10\ \mathrm{ms^{-2}}$).

(A) 20 (B) 25 (C) 30 (D) 15

Open solution

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Q30. JEE Main 2023 (Online) 31st January Morning Shift

The initial speed of a projectile fired from ground is u. At the highest point during its motion, the speed of projectile is $\frac{\sqrt{3}}{2}u$. The time of flight of the projectile is :

(A) $\frac{u}{g}$ (B) $\frac{2u}{g}$ (C) $\frac{u}{2g}$ (D) $\frac{\sqrt{3}u}{g}$

Open solution

Q31. JEE Main 2023 (Online) 29th January Morning Shift

A stone is projected at angle 30° to the horizontal. The ratio of kinetic energy of the stone at point of projection to its kinetic energy at the highest point of flight will be -

(A) 1 : 4 (B) 1 : 2 (C) 4 : 3 (D) 4 : 1

Open solution

Q32. JEE Main 2023 (Online) 25th January Evening Shift

Two objects are projected with same velocity 'u' however at different angles α and β with the horizontal. If $\alpha + \beta = 90^\circ$, the ratio of horizontal range of the first object to the 2nd object will be :

(A) 1 : 1 (B) 2 : 1 (C) 1 : 2 (D) 4 : 1

Open solution

Q33. JEE Main 2023 (Online) 24th January Morning Shift

The maximum vertical height to which a man can throw a ball is 136 m. The maximum horizontal distance upto which he can throw the same ball is :

(A) 136 m (B) 272 m (C) 68 m (D) 192 m

Open solution

Q34. JEE Main 2022 (Online) 28th July Evening Shift

At time $t = 0$ a particle starts travelling from a height $7\hat{z}\,\mathrm{cm}$ in a plane keeping z coordinate constant. At any instant of time its position along the $\hat{x}$ and $\hat{y}$ directions are defined as $3\mathrm{t}$ and $5\mathrm{t}^3$ respectively. At t = 1s acceleration of the particle will be

(A) $-30\hat{y}$ (B) $30\hat{y}$ (C) $3\hat{x} + 15\hat{y}$ (D) $3\hat{x} + 15\hat{y} + 7\hat{z}$

Open solution

Q35. JEE Main 2022 (Online) 28th July Morning Shift

An NCC parade is going at a uniform speed of $9\,\mathrm{km/h}$ under a mango tree on which a monkey is sitting at a height of $19.6\,\mathrm{m}$. At any particular instant, the monkey drops a mango. A cadet will receive the mango whose distance from the tree at time of drop is: (Given $g = 9.8\,\mathrm{m/s^2}$)

(A) 5 m (B) 10 m (C) 19.8 m (D) 24.5 m

Open solution

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Q36. JEE Main 2022 (Online) 27th July Evening Shift

A body of mass $10\,\mathrm{kg}$ is projected at an angle of 45° with the horizontal. The trajectory of the body is observed to pass through a point $(20, 10)$. If T is the time of flight, then its momentum vector, at time $\mathrm{t} = \frac{\mathrm{T}}{\sqrt{2}}$, is _____.

[Take $\mathrm{g} = 10\,\mathrm{m/s^2}$]

(A) $100\hat{i} + (100\sqrt{2} - 200)\hat{j}$

(B) $100\sqrt{2}\hat{i} + (100 - 200\sqrt{2})\hat{j}$

(C) $100\hat{i} + (100 - 200\sqrt{2})\hat{j}$

(D) $100\sqrt{2}\hat{i} + (100\sqrt{2} - 200)\hat{j}$

Open solution

Q37. JEE Main 2022 (Online) 26th July Evening Shift

Two projectiles are thrown with same initial velocity making an angle of 45° and 30° with the horizontal respectively. The ratio of their respective ranges will be :

(A) $1 : \sqrt{2}$ (B) $\sqrt{2} : 1$ (C) $2 : \sqrt{3}$ (D) $\sqrt{3} : 2$

Open solution

Q38. JEE Main 2022 (Online) 26th July Morning Shift

Two projectiles thrown at 30° and 45° with the horizontal respectively, reach the maximum height in same time. The ratio of their initial velocities is :

(A) $1 : \sqrt{2}$ (B) $2 : 1$ (C) $\sqrt{2} : 1$ (D) $1 : 2$

Open solution

Q39. JEE Main 2022 (Online) 25th July Evening Shift

A ball is projected from the ground with a speed 15 ms^{-1} at an angle θ with horizontal so that its range and maximum height are equal,
then 'tan θ' will be equal to :

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) 4

Open solution

Q40. JEE Main 2022 (Online) 30th June Morning Shift

At t = 0, truck, starting from rest, moves in the positive x-direction at uniform acceleration of 5 ms^{-2}. At t = 20 s, a ball is released from the top of the truck. The ball strikes the ground in 1 s after the release. The velocity of the ball, when it strikes the ground, will be :

(Given g = 10 ms^{-2})

(A) $100\hat{i} - 10\hat{j}$ (B) $10\hat{i} - 100\hat{j}$ (C) $100\hat{i}$ (D) $-10\hat{j}$

Open solution

Q41. JEE Main 2022 (Online) 30th June Morning Shift

Two projectiles P_1 and P_2 thrown with speed in the ratio $\sqrt{3} : \sqrt{2}$, attain the same height during their motion. If P_2 is thrown at an angle of 60° with the horizontal, the angle of projection of P_1 with horizontal will be :

(A) 15° (B) 30° (C) 45° (D) 60°

Open solution

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Q42. JEE Main 2022 (Online) 29th June Evening Shift

A person can throw a ball upto a maximum range of 100 m. How high above the ground he can throw the same ball?

(A) 25 m (B) 50 m (C) 100 m (D) 200 m

Open solution

Q43. JEE Main 2022 (Online) 27th June Morning Shift

A projectile is launched at an angle 'α' with the horizontal with a velocity 20 ms^{-1}. After 10 s, its inclination with horizontal is 'β'. The value of tanβ will be : (g = 10 ms^{-2}).

(A) tanα + 5secα (B) tanα $-$ 5secα

(C) 2tanα $-$ 5secα (D) 2tanα $+$ 5secα

Open solution

Q44. JEE Main 2022 (Online) 27th June Morning Shift

A girl standing on road holds her umbrella at 45° with the vertical to keep the rain away. If she starts running without umbrella with a speed of $15\sqrt{2}$ kmh^{-1}, the rain drops hit her head vertically. The speed of rain drops with respect to the moving girl is :

(A) 30 kmh^{-1} (B) $\frac{25}{\sqrt{2}}$ kmh^{-1}

(C) $\frac{30}{\sqrt{2}}$ kmh^{-1} (D) 25 kmh^{-1}

Open solution

Q45. JEE Main 2022 (Online) 25th June Evening Shift

Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Two identical balls A and B thrown with same velocity 'u' at two different angles with horizontal attained the same range R. If A and B reached the maximum height h_1 and h_2 respectively, then $R = 4\sqrt{h_1 h_2}$

Reason R : Product of said heights.

$h_1 h_2 = \left(\frac{u^2 \sin^2\theta}{2g}\right) \cdot \left(\frac{u^2 \cos^2\theta}{2g}\right)$

Choose the correct answer :

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Open solution

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Q46. JEE Main 2022 (Online) 24th June Morning Shift

A projectile is projected with velocity of 25 m/s at an angle θ with the horizontal. After t seconds its inclination with horizontal becomes zero. If R represents horizontal range of the projectile, the value of θ will be :

[use g = 10 m/s^2]

(A) $\frac{1}{2}\sin^{-1}\left(\frac{5t^2}{4R}\right)$

(B) $\frac{1}{2}\sin^{-1}\left(\frac{4R}{5t^2}\right)$

(C) $\tan^{-1}\left(\frac{4t^2}{5R}\right)$

(D) $\cot^{-1}\left(\frac{R}{20t^2}\right)$

Open solution

Q47. JEE Main 2021 (Online) 1st September Evening Shift

The ranges and heights for two projectiles projected with the same initial velocity at angles 42° and 48° with the horizontal are R_1, R_2 and H_1, H_2 respectively. Choose the correct option :

(A) $R_1 > R_2$ and $H_1 = H_2$

(B) $R_1 = R_2$ and $H_1 < H_2$

(C) $R_1 < R_2$ and $H_1 < H_2$

(D) $R_1 = R_2$ and $H_1 = H_2$

Open solution

Q48. JEE Main 2021 (Online) 31st August Morning Shift

A helicopter is flying horizontally with a speed 'v' at an altitude 'h' has to drop a food packet for a man on the ground. What is the distance of helicopter from the man when the food packet is dropped?

(A) $\sqrt{\frac{2ghv^2+1}{h^2}}$

(B) $\sqrt{2ghv^2+h^2}$

(C) $\sqrt{\frac{2v^2h}{g}+h^2}$

(D) $\sqrt{\frac{2gh}{v^2}+h^2}$

Open solution

Q49. JEE Main 2021 (Online) 27th August Evening Shift

A player kicks a football with an initial speed of 25 ms^{-1} at an angle of 45° from the ground. What are the maximum height and the time taken by the football to reach at the highest point during motion ? (Take g = 10 ms^{-2})

(A) h_{max} = 10 m

T = 2.5 s

(B) h_{max} = 15.625 m

T = 3.54 s

(C) h_{max} = 15.625 m

T = 1.77 s

(D) h_{max} = 3.54 m

T = 0.125 s

Open solution

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Q50. JEE Main 2021 (Online) 26th August Evening Shift

A bomb is dropped by fighter plane flying horizontally. To an observer sitting in the plane, the trajectory of the bomb is a :

(A) hyperbola

(B) parabola in the direction of motion of plane

(C) straight line vertically down the plane

(D) parabola in a direction opposite to the motion of plane

Open solution

Q51. JEE Main 2021 (Online) 20th July Morning Shift

A butterfly is flying with a velocity $4\sqrt{2}$ m/s in North-East direction. Wind is slowly blowing at 1 m/s from North to South. The resultant displacement of the butterfly in 3 seconds is :

(A) $12\sqrt{2}$ m (B) 20 m (C) 3 m (D) 15 m

Open solution

Q52. JEE Main 2021 (Online) 16th March Evening Shift

A mosquito is moving with a velocity $\vec{v} = 0.5t^2\hat{i} + 3t\hat{j} + 9\hat{k}$ m/s and accelerating in uniform conditions. What will be the direction of mosquito after 2 s?

(A) $\tan^{-1}\left(\frac{\sqrt{85}}{6}\right)$ from y-axis

(B) $\tan^{-1}\left(\frac{5}{2}\right)$ from y-axis

(C) $\tan^{-1}\left(\frac{2}{3}\right)$ from x-axis

(D) $\tan^{-1}\left(\frac{5}{2}\right)$ from x-axis

Open solution

Q53. JEE Main 2021 (Online) 26th February Evening Shift

The trajectory of a projectile in a vertical plane is y = αx $-$ βx^2, where α and β are constants and x & y are respectively the horizontal and vertical distances of the projectile from the point of projection. The angle of projection θ and the maximum height attained H are respectively given by :

(A) $\tan^{-1}\alpha, \frac{\alpha^2}{4\beta}$

(B) $\tan^{-1}\alpha, \frac{4\alpha^2}{\beta}$

(C) $\tan^{-1}\left(\frac{\beta}{\alpha}\right), \frac{\alpha^2}{\beta}$

(D) $\tan^{-1}\beta, \frac{\alpha^2}{2\beta}$

Open solution

Q54. JEE Main 2020 (Online) 6th September Evening Slot

When a car is at rest, its driver sees rain drops falling on it vertically. When driving the car with speed v, he sees that rain drops are coming at an angle 60° from the horizontal. On further increasing the speed of the car to (1 + β)v, this angle changes to 45°. The value of β is close to :

(A) 0.50 (B) 0.73 (C) 0.37 (D) 0.41

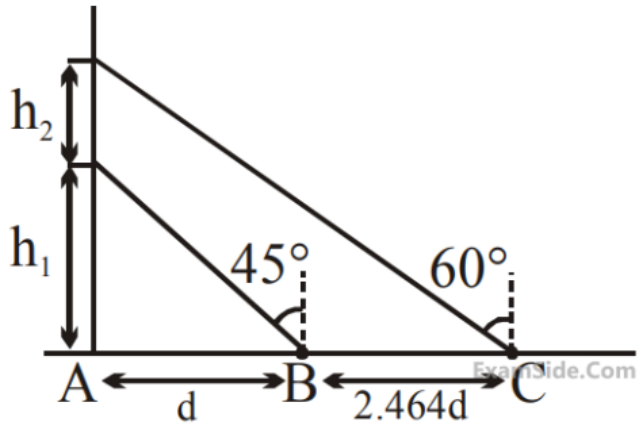
Open solution

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Q55. JEE Main 2020 (Online) 5th September Morning Slot

Open solution

A balloon is moving up in air vertically above a point A on the ground. When it is at a height h_1, a girl standing at a distance d (point B) from A (see figure) sees it at an angle 45° with respect to the vertical. When the balloon climbs up a further height h_2, it is seen at an angle 60° with respect to the vertical if the girl moves further by a distance 2.464 d (point C). Then the height h_2 is (given tan 30° = 0.5774)



(A) 0.464d (B) d (C) 0.732d (D) 1.464d

Q56. JEE Main 2020 (Online) 4th September Morning Slot

Open solution

Starting from the origin at time t = 0, with initial velocity $5\hat{j}$ ms^{-1}, a particle moves in the x-y plane with a constant acceleration of $\left(10\hat{i}+4\hat{j}\right)$ ms^{-2}. At time t, its coordinates are (20 m, y_0 m). The values of t and y_0 are, respectively:

(A) 5s and 25 m (B) 2s and 18 m (C) 2s and 24 m (D) 4s and 52 m

Q57. JEE Main 2020 (Online) 9th January Evening Slot

Open solution

A particle starts from the origin at t = 0 with an initial velocity of $3.0\,\hat{i}$ m/s and moves in the x-y plane with a constant acceleration $\left(6\hat{i}+4\hat{j}\right)$ m/s^2. The x-coordinate of the particle at the instant when its y-coordinate is 32 m is D meters. The value of D is :-

(A) 40 (B) 32 (C) 50 (D) 60

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Q58. JEE Main 2020 (Online) 8th January Evening Slot

A particle moves such that its position vector $\vec{r}(t) = \cos\omega t\hat{i} + \sin\omega t\hat{j}$ where ω is a constant and t is time. Then which of the following statements is true for the velocity $\vec{v}(t)$ and acceleration $\vec{a}(t)$ of the particle :

Open solution

(A) $\vec{v}$ and $\vec{a}$ both are perpendicular to $\vec{r}$

(B) $\vec{v}$ and $\vec{a}$ both are parallel to $\vec{r}$

(C) $\vec{v}$ is perpendicular to $\vec{r}$ and $\vec{a}$ is directed towards the origin

(D) $\vec{v}$ is perpendicular to $\vec{r}$ and $\vec{a}$ is directed away from the origin

Q59. JEE Main 2019 (Online) 12th April Evening Slot

Two particles are projected from the same point with the same speed u such that they have the same range R, but different maximum heights, h_1 and h_2. Which of the following is correct ?

Open solution

(A) $R^2 = h_1h_2$ (B) $R^2 = 16\ h_1h_2$ (C) $R^2 = 4\ h_1h_2$ (D) $R^2 = 2h_1h_2$

Q60. JEE Main 2019 (Online) 12th April Morning Slot

A shell is fired from a fixed artillery gun with an initial speed u such that it hits the target on the ground at a distance R from it. If t_1 and t_2 are the values of the time taken by it to hit the target in two possible ways, the product t_1t_2 is -

Open solution

(A) $\frac{2R}{g}$ (B) $\frac{R}{g}$ (C) $\frac{R}{2g}$ (D) $\frac{R}{4g}$

Q61. JEE Main 2019 (Online) 12th April Morning Slot

The trajectory of a projectile near the surface of the earth is given as $y = 2x - 9x^2$. If it were launched at an angle θ_0 with speed v_0 then ($g = 10\ ms^{-2}$) :

Open solution

(A) $\theta_0 = \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$ and $v_0 = \frac{5}{3}$ ms^{-1}

(B) $\theta_0 = \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$ and $v_0 = \frac{3}{5}$ ms^{-1}

(C) $\theta_0 = \sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ and $v_0 = \frac{3}{5}$ ms^{-1}

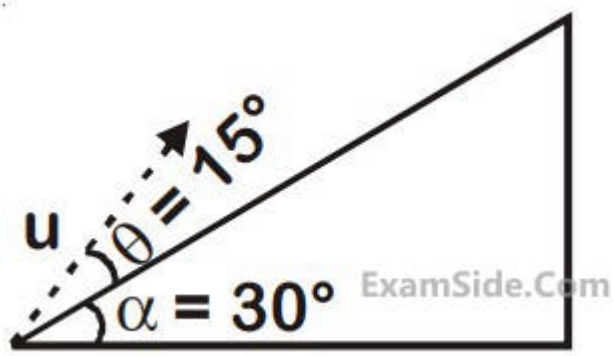
(D) $\theta_0 = \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$ and $v_0 = \frac{5}{3}$ ms^{-1}

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Q62. JEE Main 2019 (Online) 10th April Evening Slot

Open solution

A plane is inclined at an angle α = 30° with respect to the horizontal. A particle is projected with a speed u = 2 ms^{-1} , from the base of the plane, making an angle θ = 15° with respect to the plane as shown in the figure. The distance from the base, at which the particle hits the plane is close to : (Take g = 10 ms^{-2})



(A) 14 cm (B) 18 cm (C) 20 cm (D) 26 cm

Q63. JEE Main 2019 (Online) 9th April Morning Slot

Open solution

The stream of a river is flowing with a speed of 2km/h. A swimmer can swim at a speed of 4km/h. What should be the direction of the swimmer with respect to the flow of the river to cross the river straight ?

(A) 150° (B) 120° (C) 60° (D) 90°

Q64. JEE Main 2019 (Online) 8th April Morning Slot

Open solution

Ship A is sailing towards north-east with velocity $\vec{v} = 30\hat{i} + 50\hat{j}$ km/hr where $\hat{i}$ points east and $\hat{j}$, north. Ship B is at a distance of 80 km east and 150 km north of Ship A and is sailing towards west at 10 km/hr. A will be at minimum distance from B in :

(A) 2.2 hrs (B) 4.2 hrs (C) 2.6 hrs (D) 3.2 hrs

Q65. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

A person standing on an open ground hears the sound of a jet aeroplane, coming from north at an angle 60° with ground level. But he finds the aeroplane right vertically above his position. If v is the speed of sound, speed of the plane is :

(A) $\frac{\sqrt{3}}{2}$v (B) $\frac{2v}{\sqrt{3}}$ (C) v (D) $\frac{v}{2}$

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Q66. JEE Main 2019 (Online) 10th January Morning Slot

Two guns A and B can fire bullets at speeds 1 km/s and 2 km/s respectively. From a point on a horizontal ground, they are fired in all possible directions. The ratio of maximum areas covered by the bullets fired by the two guns, on the ground is -

Open solution

(A) 1 : 16 (B) 1 : 8 (C) 1 : 2 (D) 1 : 4

Q67. JEE Main 2019 (Online) 9th January Evening Slot

The position co-ordinates of a particle moving in a 3-D coordinate system is given by
x = a cosωt
y = a sinωt and
z = aωt

Open solution

The speed of the particle is :

(A) $\sqrt{2}\,a\omega$ (B) $a\omega$ (C) $\sqrt{3}\,a\omega$ (D) 2aω

Q68. JEE Main 2019 (Online) 9th January Morning Slot

A particle is moving with a velocity

$\overrightarrow{v} = K(y\hat{i} + x\hat{j})$, where K is a constant.

Open solution

The general equation for its path is :

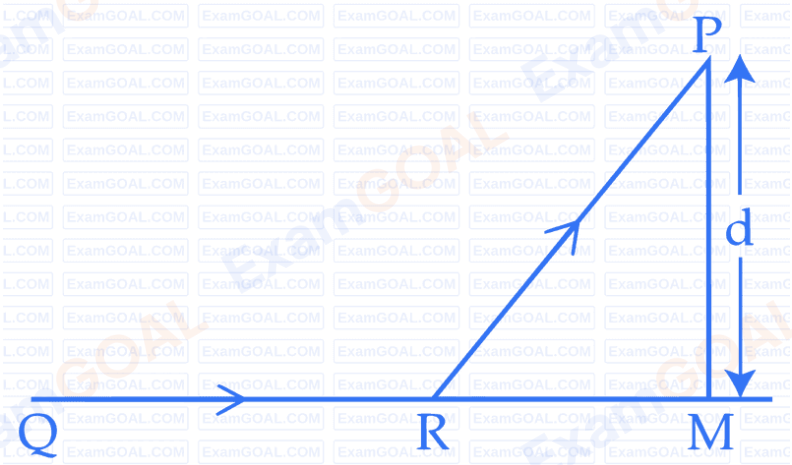
(A) $y = x^2$ + constant (B) $y^2 = x$ + constant
(C) $y^2 = x^2$ + constant (D) xy = constant

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Q69. JEE Main 2018 (Online) 15th April Evening Slot

A man in a car at location Q on a straight highway is moving with speed v. He decides to reach a point P in a field at a distance d from the highway (point M) as shown in the figure. Speed of the car in the field is half to that on the highway. What should be the distance RM, so that the time taken to reach P is minimum ?

Open solution



(A) d (B) $\frac{d}{\sqrt{2}}$ (C) $\frac{d}{2}$ (D) $\frac{d}{\sqrt{3}}$

Q70. JEE Main 2014 (Online) 11th April Morning Slot

The initial speed of a bullet fired from a rifle is 630 m/s. The rifle is fired at the center of a target 700 m away at the same level as the target. How far above the center of the target the rifle must be aimed in order to hit the target ?

Open solution

(A) 1.0 m (B) 4.2 m (C) 6.1 m (D) 9.8 m

Q71. JEE Main 2014 (Online) 9th April Morning Slot

The position of a projectile launched from the origin at t = 0 is given by $\vec{r}$ = (40 $\hat{i}$ + 50 $\hat{j}$) m at t = 2s. If the projectile was launched at an angle θ from the horizontal, then θ is (take g = 10 ms^{-2}).

Open solution

(A) $\tan^{-1}\frac{2}{3}$ (B) $\tan^{-1}\frac{3}{2}$ (C) $\tan^{-1}\frac{7}{4}$ (D) $\tan^{-1}\frac{4}{5}$

Q72. JEE Main 2013 (Online) 25th April Morning Slot

The maximum range of a bullet fired from a toy pistol mounted on a car at rest is $R_0 = 40$ m. What will be the acute angle of inclination of the pistol for maximum range when the car is moving in the direction of firing with uniform velocity $v = 20$ m/s on a horizontal surface? ($g = 10$ m/s^2)

Open solution

(A) 30° (B) 60° (C) 75° (D) 45°

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Q73. JEE Main 2013 (Online) 22th April Morning Slot

A ball projected from ground at an angle of 45° just clears a wall in front. If point of projection is 4 m from the foot of wall and ball strikes the ground at a distance of 6 m on the other side of the wall, the height of the wall is :

Open solution

(A) 4.4 m (B) 2.4 m (C) 3.6 m (D) 1.6 m

Q74. JEE Main 2013 (Offline)

A projectile is given an initial velocity of $\left(\hat{i}+2\hat{j}\right)$ m/s, where $\hat{i}$ is along the ground and $\hat{j}$ is along the vertical. If g = 10 m/s^2, the equation of its trajectory is:

Open solution

(A) y = x - 5x^2 (B) y = 2x - 5x^2 (C) 4y = 2x - 5x^2 (D) 4y = 2x - 25x^2

Q75. AIEEE 2012

A boy can throw a stone up to a maximum height of 10 m. The maximum horizontal distance that the boy can throw the same stone up to will be

Open solution

(A) $20\sqrt{2}$ m (B) 10 m (C) $10\sqrt{2}$ m (D) 20 m

Q76. AIEEE 2011

A water fountain on the ground sprinkles water all around it. If the speed of water coming out of the fountain is v, the total area around the fountain that gets wet is :

Open solution

(A) $\pi\frac{v^4}{g^2}$ (B) $\frac{\pi}{2}\frac{v^4}{g^2}$

(C) $\pi\frac{v^2}{g^2}$ (D) $\pi\frac{v^2}{g}$

Q77. AIEEE 2010

A particle is moving with velocity $\overrightarrow{v}=k\left(y\hat{i}+x\hat{j}\right)$, where K is a constant. The general equation for its path is

Open solution

(A) y = x^2 + constant (B) y^2 = x + constant

(C) xy = constant (D) y^2 = x^2 + constant

Q78. AIEEE 2009

A particle has an initial velocity $3\hat{i}+4\hat{j}$ and an acceleration of $0.4\hat{i}+0.3\hat{j}$. Its speed after 10 s is:

Open solution

(A) $7\sqrt{2}$ units (B) 7 units (C) 8.5 units (D) 10 units

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Q79. AIEEE 2005

A particle is moving eastwards with a velocity of 5 m/s. In 10 seconds the velocity changes to 5 m/s northwards. The average acceleration in this time is

(A) $\frac{1}{2}ms^{-2}$ towards north

(B) $\frac{1}{\sqrt{2}}ms^{-2}$ towards north-east

(C) $\frac{1}{\sqrt{2}}ms^{-2}$ towards north-west

(D) zero

Open solution

Q80. AIEEE 2005

A projectile can have the same range R for two angles of projection. If t_1 and t_2 be the times of flights in the two cases, then the product of the two time of flights is proportional to

(A) R^2

(B) $\frac{1}{R^2}$

(C) $\frac{1}{R}$

(D) R

Open solution

Q81. AIEEE 2004

A projectile can have the same range 'R' for two angles of projection. If T_1 and T_2 be the time of flights in the two cases, then the product of the two time of flights is directly proportional to

(A) R

(B) $\frac{1}{R}$

(C) $\frac{1}{R^2}$

(D) R^2

Open solution

Q82. AIEEE 2004

A ball is thrown from a point with a speed v_0 at an angle of projection θ. From the same point and at the same instant person starts running with a constant speed $\frac{v_0}{2}$ to catch the ball. Will the person be able to catch the ball? If yes, what should be the angle of projection θ?

(A) No

(B) Yes, 30°

(C) Yes, 60°

(D) Yes, 45°

Open solution

Q83. AIEEE 2003

A boy playing on the roof of a 10 m high building throws a ball with a speed of 10 m/s at an angle of 30° with the horizontal. How far from the throwing point will the ball be at the height of 10 m from the ground? $\left[g = 10m/s^2, \sin 30^\circ = \frac{1}{2}, \cos 30^\circ = \frac{\sqrt{3}}{2}\right]$

(A) 5.20 m

(B) 4.33 m

(C) 2.60 m

(D) 8.66 m

Open solution

Numerical 20 questions

Q84. JEE Main 2026 (Online) 4th April Evening Shift

A gun mounted on the ground fires bullets in all directions with same speed. The farthest distance the bullets could reach is 6.4 m . The speed of the bullets from the gun is ____ $\mathrm{m/s}$.

(take $\mathrm{g = 10\ m/s^2}$)

Open solution

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Q85. JEE Main 2025 (Online) 29th January Morning Shift

The maximum speed of a boat in still water is 27 km/h. Now this boat is moving downstream in a river flowing at 9 km/h. A man in the boat throws a ball vertically upwards with speed of 10 m/s. Range of the ball as observed by an observer at rest on the bank is _____ cm. (Take $g = 10$ m/s^2)

Open solution

Q86. JEE Main 2025 (Online) 22nd January Morning Shift

A particle is projected at an angle of 30° from horizontal at a speed of $60\ \mathrm{m/s}$. The height traversed by the particle in the first second is $\mathrm{h_0}$ and height traversed in the last second, before it reaches the maximum height, is h_1. The ratio $h_0 : h_1$ is _____.

[Take, $\mathrm{g} = 10\ \mathrm{m/s^2}$]

Open solution

Q87. JEE Main 2024 (Online) 8th April Evening Shift

A body of mass M thrown horizontally with velocity v from the top of the tower of height H touches the ground at a distance of $100\ \mathrm{m}$ from the foot of the tower. A body of mass $2\ \mathrm{M}$ thrown at a velocity $\frac{v}{2}$ from the top of the tower of height $4\mathrm{H}$ will touch the ground at a distance of _____ m.

Open solution

Q88. JEE Main 2024 (Online) 5th April Evening Shift

The maximum height reached by a projectile is $64\ \mathrm{m}$. If the initial velocity is halved, the new maximum height of the projectile is _____ m.

Open solution

Q89. JEE Main 2024 (Online) 29th January Morning Shift

A ball rolls off the top of a stairway with horizontal velocity u. The steps are $0.1\ \mathrm{m}$ high and $0.1\ \mathrm{m}$ wide. The minimum velocity u with which that ball just hits the step 5 of the stairway will be $\sqrt{x}\ \mathrm{ms}^{-1}$ where $x =$ _____ [use $\mathrm{g} = 10\ \mathrm{m/s^2}$].

Open solution

Q90. JEE Main 2024 (Online) 27th January Morning Shift

A particle starts from origin at $t = 0$ with a velocity $5\hat{i}\ \mathrm{m/s}$ and moves in $x - y$ plane under action of a force which produces a constant acceleration of $(3\hat{i} + 2\hat{j})\mathrm{m/s^2}$. If the x-coordinate of the particle at that instant is $84\ \mathrm{m}$, then the speed of the particle at this time is $\sqrt{\alpha}\ \mathrm{m/s}$. The value of α is _____.

Open solution

Q91. JEE Main 2023 (Online) 11th April Morning Shift

A projectile fired at 30° to the ground is observed to be at same height at time $3\ \mathrm{s}$ and $5\ \mathrm{s}$ after projection, during its flight. The speed of projection of the projectile is _____ $\mathrm{m\ s^{-1}}$.

(Given $g = 10\ \mathrm{ms}^{-2}$)

Open solution

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Q92. JEE Main 2023 (Online) 31st January Evening Shift

Two bodies are projected from ground with same speeds $40\ \mathrm{ms}^{-1}$ at two different angles with respect to horizontal. The bodies were found to have same range. If one of the body was projected at an angle of 60°, with horizontal then sum of the maximum heights, attained by the two projectiles, is _____ m. (Given $\mathrm{g} = 10\ \mathrm{ms}^{-2}$)

Open solution

Q93. JEE Main 2023 (Online) 31st January Morning Shift

The speed of a swimmer is $4\ \mathrm{km\ h}^{-1}$ in still water. If the swimmer makes his strokes normal to the flow of river of width $1\ \mathrm{km}$, he reaches a point $750\ \mathrm{m}$ down the stream on the opposite bank.

The speed of the river water is _____ $\mathrm{km\ h}^{-1}$

Open solution

Q94. JEE Main 2022 (Online) 30th July Evening Shift

Two projectiles are thrown towards each other at 15° and 45° angles respectively with the horizontal at the same speed. The difference between the horizontal distances traveled by the two projectiles is $80\ \mathrm{m}$. The initial speed of the projectiles is _____ $\mathrm{m\,s}^{-1}$

[Given $g = 10\ \mathrm{m\,s}^{-2}$]

Open solution

Q95. JEE Main 2022 (Online) 29th July Morning Shift

An object is projected in the air with initial velocity u at an angle θ. The projectile motion is such that the horizontal range R, is maximum. Another object is projected in the air with a horizontal range half of the range of first object. The initial velocity remains same in both the cases. The value of the angle of projection, at which the second object is projected, will be _____ degree.

Open solution

Q96. JEE Main 2022 (Online) 27th July Morning Shift

A ball of mass m is thrown vertically upward. Another ball of mass $2\ \mathrm{m}$ is thrown at an angle θ with the vertical. Both the balls stay in air for the same period of time. The ratio of the heights attained by the two balls respectively is $\frac{1}{x}$. The value of x is _____.

Open solution

Q97. JEE Main 2022 (Online) 26th July Morning Shift

If the initial velocity in horizontal direction of a projectile is unit vector $\hat{i}$ and the equation of trajectory is $y = 5x(1 - x)$. The y component vector of the initial velocity is _____ $\hat{j}$. $(\mathrm{Take\ g} = 10\ \mathrm{m/s}^2)$

Open solution

Q98. JEE Main 2022 (Online) 26th June Morning Shift

A fighter jet is flying horizontally at a certain altitude with a speed of 200 ms^{-1}. When it passes directly overhead an anti-aircraft gun, a bullet is fired from the gun, at an angle θ with the horizontal, to hit the jet. If the bullet speed is 400 m/s, the value of θ will be _____$^{\circ}$.

Open solution

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Q99. JEE Main 2022 (Online) 24th June Evening Shift

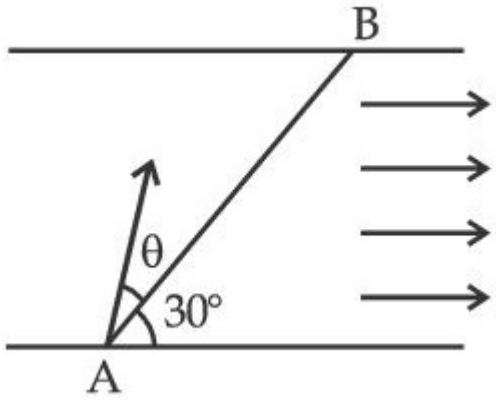
A body is projected from the ground at an angle of 45° with the horizontal. Its velocity after 2s is 20 ms^{-1}. The maximum height reached by the body during its motion is _____ m. (use g = 10 ms^{-2})

Open solution

Q100. JEE Main 2021 (Online) 27th July Evening Shift

A swimmer wants to cross a river from point A to point B. Line AB makes an angle of 30° with the flow of river. Magnitude of velocity of the swimmer is same as that of the river. The angle θ with the line AB should be _____°, so that the swimmer reaches point B.

Open solution



Q101. JEE Main 2021 (Online) 18th March Morning Shift

A person is swimming with a speed of 10 m/s at an angle of 120° with the flow and reaches to a point directly opposite on the other side of the river. The speed of the flow is 'x' m/s. The value of 'x' to the nearest integer is _____.

Open solution

Q102. JEE Main 2021 (Online) 16th March Evening Shift

A swimmer can swim with velocity of 12 km/h in still water. Water flowing in a river has velocity 6 km/h. The direction with respect to the direction of flow of river water he should swim in order to reach the point on the other bank just opposite to his starting point is _____°. (Round off to the Nearest Integer) (Find the angle in degrees)

Open solution

Q103. JEE Main 2020 (Online) 8th January Morning Slot

A particle is moving along the x-axis with its coordinate with the time 't' given be $x(t) = 10 + 8t - 3t^2$. Another particle is moving the y-axis with its coordinate as a function of time given by $y(t) = 5 - 8t^3$.
At t = 1s, the speed of the second particle as measured in the frame of the first particle is given as $\sqrt{v}$. Then v (in m/s) is _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **B**	3. **B**	4. **D**	5. **B**
6. **C**	7. **C**	8. **C**	9. **A**	10. **D**
11. **D**	12. **A**	13. **D**	14. **A**	15. **C**
16. **C**	17. **D**	18. **D**	19. **A**	20. *Scan QR*
21. **D**	22. **A**	23. **C**	24. **B**	25. **B**
26. **A**	27. **A**	28. **A**	29. **D**	30. **A**
31. **C**	32. **A**	33. **B**	34. **B**	35. **A**
36. **D**	37. **C**	38. **C**	39. **D**	40. **A**
41. **C**	42. **B**	43. **B**	44. **C**	45. **A**
46. **D**	47. **B**	48. **C**	49. **C**	50. **C**
51. **D**	52. **A**	53. **A**	54. **B**	55. **B**
56. **B**	57. **D**	58. **C**	59. **B**	60. **A**
61. **A**	62. **C**	63. **B**	64. **C**	65. **D**
66. **A**	67. **A**	68. **C**	69. **D**	70. **C**
71. **C**	72. **B**	73. **B**	74. **B**	75. **D**
76. **A**	77. **D**	78. **A**	79. **C**	80. **D**
81. **A**	82. **C**	83. **D**		

Numerical

84. **8**	85. **2000**	86. **5**	87. **100**	88. **16**
89. **2**	90. **673**	91. **80**	92. **80**	93. **3**
94. **40**	95. **15 or 75**	96. **1**	97. **5**	98. **60**
99. **20**	100. **30**	101. **5**	102. **120**	103. **580**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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ANALYSIS

Know your score. Know your rank.

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- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

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